

Dispatch — Emergency Response Routing Sim

Group Rapid Route — Assignment 2: Project Approval, Specification, and MVP Definition

Product Requirement Document (PRD)

Problem & Opportunity

Emergency dispatch only matters against a believable backdrop of ordinary city traffic — and most simulations fake that backdrop with randomly assigned congestion rather than simulating it. We want to build the traffic layer the way transportation engineers actually model it (demand generated by zone density, distributed by an origin-destination matrix, assigned to routes by expected cost, and realized by car-following agents that obey basic traffic laws at junctions), and then show emergency response operating inside that realistic substrate rather than a hand-waved one.

Target Audience

Computer graphics / visual computing students and instructors evaluating crowd-simulation and wayfinding technique; peers interested in agent-based animation, traffic microsimulation, or emergency-response modeling; secondarily, anyone curious about how real city traffic models (OD matrices, trip generation, route assignment) translate into a visual simulation.

User Stories

- As a viewer, I want denser parts of the city (taller buildings, more zones) to generate more traffic so the map's traffic pattern reflects the city's actual layout.
- As a viewer, I want traffic to follow realistic origin-destination flows (e.g. residential → downtown in the morning) rather than spawning at uniformly random points.
- As a viewer, I want to see traffic get heavier at simulated peak hours so the city feels alive on a daily cycle.
- As a viewer, I want cars to keep a safe following distance and never overlap or visually desync (no “ghost cars”) so the traffic reads as real, not glitchy.
- As a viewer, I want cars approaching a shared junction to take turns by basic right-of-way rules, not collide or freeze.
- As a user, I want the routing algorithm to choose the path with the lowest expected travel cost, not just the shortest distance, so a longer-but-faster route can be correctly preferred.
- As a viewer, I want to see incidents spawn on the live map and emergency units dispatched to them, moving through the same realistic traffic as everyone else.
- As a user, I want to trigger a road closure or manually spawn an incident so I can test how both civilian traffic and emergency routing react to disruption.
- As a user, I want to see aggregate stats (response time, calls handled) so I can evaluate system performance.

Key Features (MoSCoW)

Must have	Zone-based city map with a building-density value per zone; an OD (origin-destination) matrix that drives trip generation and distribution, including peak vs. off-peak variation; car-following traffic microsimulation that maintains safe following distance and produces no overlapping or desynced (“ghost”) cars; junction-level conflict resolution (basic right-of-way rules) so cars never collide or freeze at intersections; a cost-based route assignment algorithm that weighs expected travel time, not raw distance, so it can prefer a longer but faster route.
Should have	Emergency incident spawning and unit dispatch layered on top of the traffic substrate (nearest-available matching unit, live rerouting around closures); mutual-aid fallback; response-time and calls-handled tracking; a fleet/incident status UI.
Could have	Multi-unit incidents requiring more than one responder; severity-weighted dispatch; adaptive traffic-signal timing; a non-grid, procedurally generated road network; camera pan/zoom; 3D rendering polish.
Won't have (this semester)	Multi-modal transport (pedestrians, transit, cyclists); real-world GIS map import; machine-learned demand prediction; multiplayer/multi-user control.

Technical Specification

Tech Stack

- **Framework:** Browser-based, custom lightweight 2D engine on HTML5 Canvas — no external game engine needed for the grid-based scope. Three.js is the fallback if the project extends to 3D later.

- **Language:** JavaScript for the MVP; may move to TypeScript as the codebase grows.
- **External libraries:** None required for the MVP (traffic microsimulation, routing, and rendering are custom). Candidates for later: Chart.js or D3.js for a metrics dashboard.

AI Integration Strategy

Development workflow (offline): Using Claude/ChatGPT to scaffold the traffic model (zone/density data structures, OD matrix generation, car-following and junction logic, cost-based route assignment), iterate on the visual layout, and generate concept mockups before implementation.

Runtime integration (online): The techniques driving this project are classical, agent-based models that run entirely client-side: an OD-matrix-driven demand model, a car-following microsimulation (a lightweight Intelligent-Driver-Model-style rule: speed depends on the gap to the car ahead) with junction-level right-of-way rules, and a cost-based (time-dependent) shortest-path algorithm for route assignment. This sits squarely in the course's visual-computing scope (crowd simulation, wayfinding) rather than being a general-purpose algorithms exercise. No LLM APIs are used at runtime.

Performance Constraints

- **Target platform:** Web browser, desktop-first with a responsive layout for smaller screens.
- **Target frame rate:** 60 FPS with a city-scale traffic population (target: a few hundred concurrent vehicles) plus up to ~50 emergency units and ~200 road segments.

MVP Definition (Narrow Vertical Slice)

Single most technically difficult feature: A working traffic microsimulation core: zone-based demand (density-driven trip generation, OD-matrix-driven distribution, peak/off-peak variation) realized by car-following agents that maintain safe following distance, never overlap or desync, and resolve conflicts at shared junctions by basic right-of-way rules, combined with a cost-based route assignment that picks the lowest expected-travel-time path rather than the shortest-distance one. This is deliberately reordered from our original plan: the traffic substrate is now the hard problem, and emergency dispatch is a layer we add once it is solid, since dispatch only means something against believable ambient traffic.

How we will prove it by the mid-term: A demo where (1) trips are generated from zone density and distributed by an OD matrix, visibly producing heavier flow toward high-density zones and at a simulated peak hour; (2) a few hundred cars move through the network maintaining safe following distances with no visible overlaps, collisions, or desynced positions at any junction; and (3) at least one example route is shown choosing a longer, less-congested path over a shorter, jammed one, with the two candidate costs logged side by side to make the trade-off explicit.

What is explicitly cut from the MVP:

- Emergency incident spawning and dispatch — explicitly deferred to a Should-have layer added once the traffic core is validated, not part of the MVP itself.
- A physically accurate driver-behavior model (full Intelligent Driver Model with calibrated parameters, lane-changing, acceleration curves) — MVP uses a simplified car-following rule that looks convincing.
- Adaptive or learned traffic-signal timing — MVP uses fixed right-of-way rules at junctions.
- Multi-modal transport (pedestrians, transit, cyclists) and real-world GIS map import.
- Visual polish (city blocks, motion trails, glow effects) beyond what's needed to read the simulation clearly.

Interactive Mockup (Proof of Logic)

Our existing 3D prototype (Three.js) already implements A* routing, live congestion, and road closures with rerouting for a small emergency fleet, and doubles as a proof-of-logic mockup for the routing half of this system. It predates this revision and does not yet include the OD-matrix-driven traffic core, car-following microsimulation, or cost-based route assignment described above — those are the next things we are building toward the mid-term:

<https://claude.ai/artifact/YHhhN9f618rFX2LeG3Dicb>

Proof of Approval & Mentor Assignment

[CS 549] Project Approval Request - Group Vector Response External Inbox x



Mihir Umesh Shinde <mus12@scarletmail.rutgers.edu>
to mk1353, mubbasir.kapadia, DANRUI, Sen, Ananya

Sep 21, 2026, 4:52 PM (8 days ago) ☆ 😊 ↩ ⋮

Dear Mubbasir Kapadia, Danrui, and Sen,

Our group (Rapid Route) would like to request formal approval for our semester project concept, revised based on TA feedback.

Project summary: "Dispatch" is an interactive visual simulation of a city's traffic and emergency response. The core of the project is a realistic traffic substrate: a zone-based city map where building density drives trip generation, an origin-destination (OD) matrix distributes trips across zones with peak/off-peak variation, and a car-following microsimulation with junction-level right-of-way rules moves hundreds of vehicles without overlaps or desyncs. Routes are assigned by lowest expected travel cost, not raw distance. Emergency units (ambulances, fire trucks, police) are then dispatched through this same realistic traffic, rather than through hand-waved random congestion.

Selected pillars (changed):

1. Pathfinding & Navigation
2. Swarm & Boids Modeling

Could you please confirm formal approval of this revised concept and let us know which mentor will be assigned to our group for the semester?

Thank you,

Mihir Shinde (Group Leader)
On behalf of Group Rapid Route



danrui.li@rutgers.edu
to me, mk1353, mubbasir.kapadia, DANRUI, Sen, Ananya

Sep 21, 2026, 5:01 PM (8 days ago) ☆ 😊 ↩ ⋮

Hi Mihir,

Your proposal is approved. Your project will be assigned to me (Danrui).

Sincerely,
Danrui Li